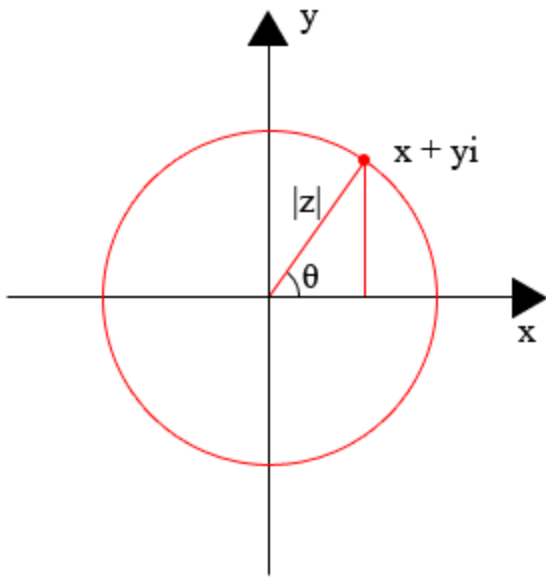


The complex plane



The polar form of a complex number

$$z = r(\cos(\theta) + i\sin(\theta))$$

DeMoivre's Theorem:

$$\text{if } z = r(\cos(\theta) + i\sin(\theta))$$

$$\text{then } z^n = r^n(\cos(n\theta) + i\sin(n\theta))$$

Example:

$$z = (1 + i) = 2^{1/2}(\cos(\pi/4) + i\sin(\pi/4))$$

$$z^{1/2} = 2^{1/4}(\cos(\pi/8) + i\sin(\pi/8)) =$$

$$2^{1/4}(0.9238 + 0.3826i) = 1.0985 + 0.4549i$$